Ordinary generating functions: formal power series, recurrences, Fibonacci, Catalan

Notes from Carlos' lessons

Source lessons: 05-09, 05-21 (Fibonacci), 05-27 (Catalan), 06-06 (Catalan, cont.), and the recurrence example on the left of 06-13.

1 Formal power series (05-09, 05-21)

Goals (05-09): find closed forms for recurrences; even without a closed form, read off asymptotics. Convergence is a nuisance, so treat a power series **formally**: an infinite sequence $(a_n)_{n\geq 0} = (a_0, a_1, \dots)$ written $\sum a_n x^n$.

- Sum: $(a_n) + (b_n) = (a_n + b_n)$. Product (convolution): $(a_n)(b_n) = \left(\sum_{k=0}^{n} a_k b_{n-k}\right)_n = (a_0 b_0, a_1 b_0 + a_0 b_1, \dots)$. Identities: $0 = (0, 0, \dots)$ and $1 = (1, 0, 0, \dots)$. The series x^n is $(0, \dots, 0, 1, 0, \dots)$ with the 1 in position n; the constant k is $(k, 0, 0, \dots)$. These form the ring $\mathbb{R}[[x]]$. Sequences that are eventually 0 behave like polynomials.
- Combinatorial reading: if A counts structures "a" of size n and B counts structures "b" of size n, then $A + B$ counts "a or b" and AB counts pairs (a, b) with total size n.

Geometric series. $(1 - x)(1 + x + x^2 + \cdots) = (1, -1, 0, \dots)(1, 1, 1, \dots) = (1, 0, 0, \dots) = 1$. So $1 + x + x^2 + \cdots = \dfrac{1}{1 - x}$ (meaning: it is the inverse of $1 - x$ in the ring). Replacing $x \to tx$: $\dfrac{1}{1 - tx} = (1, t, t^2, \dots)$.

Inverses.

Claim 1. *(a_n) has a multiplicative inverse iff $a_0 \neq 0$; the inverse is then unique.*

Proof. We want b with $\sum_{k\leq n} a_k b_{n-k} = [n = 0]$. For $n = 0$: $a_0 b_0 = 1$, which needs $a_0 \neq 0$ (and $b_0 = 1/a_0$). For $n \geq 1$: $b_n = -\frac{1}{a_0} \sum_{k=1}^{n} a_k b_{n-k}$ is forced, so b exists and is unique. Conversely, if $a_0 = 0$ then every product has 0-th term $0 \neq 1$. □

Hence $1/x$ is *not* defined ($x = (0, 1, 0, \dots)$ has $a_0 = 0$), as the board notes.

2 Solving a recurrence: $a_0 = 1,\ a_{n+1} = 2a_n + n$

(Left of the 06-13 board.) Let $A(x) = \sum a_n x^n$. Multiply the recurrence by x^{n+1} and sum over $n \geq 0$:

$$A(x) - 1 = 2xA(x) + \sum_{n\geq 0} n x^{n+1} = 2xA(x) + \frac{x^2}{(1-x)^2},$$

using $\sum n x^n = \frac{x}{(1-x)^2}$ (the sequence $(0, 1, 2, \dots)$ on the board). So $A(x) = \dfrac{1}{1-2x} + \dfrac{x^2}{(1-2x)(1-x)^2}$. Partial fractions give

$$A(x) = \frac{2}{1-2x} - \frac{1}{1-x} - \frac{x}{(1-x)^2},$$

and reading off coefficients,

$$a_n = 2^{n+1} - 1 - n.$$

Check: $a = 1, 2, 5, 12, \dots$ and the formula gives $1, 2, 5, 12$. (The board's final line is $2^{n+1} - n - 1$.)

The 05-27 board does a simpler version: $\dfrac{1}{(1-x)(1-2x)} = \dfrac{-1}{1-x} + \dfrac{2}{1-2x}$ (solving $A + B = 1$, $-2A - B = 0$ gives $A = -1$, $B = 2$), so the coefficients are $2^{n+1} - 1$. That is the solution of $T_n = 2T_{n-1} + 1$ with $T_0 = 1$ (I could not tell whether the board's T_0 was 0 or 1).

3 Fibonacci (05-21)

$F_0 = 0$, $F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$; let $f = (F_0, F_1, \dots)$. The board's key observation: multiplying by x^n shifts the sequence right by n places.

$$f = (0, 1, 1, 2, \dots), \quad xf = (0, 0, 1, 1, \dots), \quad x^2 f = (0, 0, 0, 1, \dots),$$
$$\Rightarrow\ (x + x^2)f = (0, 0, F_2, F_3, \dots) = f - x,$$

because $F_{n+1} = F_n + F_{n-1}$ makes $(x+x^2)f$ agree with f from index 2 onward, while f has $F_0 = 0$, $F_1 = 1$ at the front. Hence $x = (1 - x - x^2)f$ and

$$\boxed{f = \frac{x}{1 - x - x^2}}.$$

Closed form. $1 - x - x^2 = (1 - \varphi x)(1 - \psi x)$ with $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$ (so $\varphi + \psi = 1$ and $\varphi\psi = -1$; these are $-1/r_{1,2}$ for the roots $r_{1,2} = \frac{-1 \mp \sqrt{5}}{2}$ of $1 - x - x^2$, as the board writes with $r_1 = -\varphi$, $r_2 = -\varphi^{-1}$). Partial fractions:

$$\frac{x}{(1-\varphi x)(1-\psi x)} = \frac{1}{\sqrt{5}}\Big(\frac{1}{1-\varphi x} - \frac{1}{1-\psi x}\Big),$$

and $\frac{1}{1-tx} = (1, t, t^2, \dots)$ (the board's "final step"), so

$$\boxed{F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}}.$$

(The board does the partial fractions in the form $\frac{1}{(x-r_1)(x-r_2)} = \frac{A}{x-r_1} + \frac{B}{x-r_2}$ with $A = -\frac{1}{\sqrt{5}}$, $B = \frac{1}{\sqrt{5}}$, and rewrites $\frac{1}{x-t} = -\frac{1}{t} \cdot \frac{1}{1-x/t}$.) Verified numerically for $n < 20$.

4 Catalan numbers (05-27, 06-06)

Recurrence. $C_0 = 1$ and $C_{n+1} = \sum_{k=0}^{n} C_k C_{n-k}$ for $n \geq 0$. Let $C(x) = \sum C_n x^n$ and write $[x^n]\,A$ for the n-th coefficient. Coefficient rules from the board: $[x^n]\,(A+B) = [x^n]\,A + [x^n]\,B$; $[x^n]\,(AB) = \sum_{k=0}^{n} [x^k]\,A \cdot [x^{n-k}]\,B$; and $[x^n]\,(xP) = [x^{n-1}]\,P$.

Functional equation. For $n \geq 0$: $[x^{n+1}]\,C = \sum_k C_k C_{n-k} = [x^n]\,C^2 = [x^{n+1}]\,(xC^2)$, and $[x^0]\,C = 1$. Therefore

$$C(x) = 1 + x\,C(x)^2, \qquad \text{i.e.} \qquad xC^2 - C + 1 = 0.$$

Solving. The quadratic formula (in the region where the series converges) gives $C(x) = \dfrac{1 \pm \sqrt{1-4x}}{2x}$. Which sign? Since $C(0) = C_0 = 1$, i.e. $\lim_{x\to 0} C(x) = 1$ (the continuity remark on the 06-06 board: "$C(x)$ must converge to a continuous function wherever it converges"):

- "+": $\dfrac{1+\sqrt{1-4x}}{2x} \to \pm\infty$ as $x \to 0^+$. Rejected.
- "−": $\dfrac{1-\sqrt{1-4x}}{2x} = \dfrac{4x}{2x\,(1+\sqrt{1-4x})} \to 1$. Accepted.

So $C(x) = \dfrac{1-\sqrt{1-4x}}{2x}$.

Coefficients via the binomial series. For any real r, $(1+x)^r = \sum_{k\geq 0} \binom{r}{k} x^k$ with $\binom{r}{k} = \dfrac{r(r-1)\cdots(r-k+1)}{k!}$. (The board motivates this as: for $n \in \mathbb{N}$ it is the usual binomial theorem, and the falling-factorial form makes sense for $r = \frac{1}{2}$, "a power series for $\sqrt{1+x}$".) Then

$$\sqrt{1-4x} = \sum_{k\geq 0} \binom{1/2}{k} (-4x)^k, \qquad C(x) = -\frac{1}{2x} \sum_{k\geq 1} \binom{1/2}{k} (-4)^k x^k = \sum_{k\geq 0} 2\binom{1/2}{k+1} (-4)^k x^k,$$

the boxed expression on the board. Simplifying (my step; the board stops here) gives $C_n = \dfrac{1}{n+1}\binom{2n}{n}$. For example $n = 1$: $2\binom{1/2}{2}(-4) = 2 \cdot (-\frac{1}{8})(-4) = 1$; and $n = 2$: $2 \cdot \frac{1}{16} \cdot 16 = 2$. Numerically $1, 1, 2, 5, 14, 42, 132, 429, \ldots$.

Convergence remarks (06-06). The series $\sum C_n x^n$ cannot converge for $x > \frac{1}{4}$ because $\sqrt{1-4x}$ is non-real there; the radius of convergence is $\frac{1}{4}$. The board also writes out $C(\frac{1}{4}) = C_0 + \frac{1}{4}C_1 + \frac{1}{16}C_2 + \cdots > 1$ while probing the boundary of the region. (Independently, $C_n \sim 4^n/(\sqrt{\pi}\, n^{3/2})$ confirms the radius $\frac{1}{4}$.) The trailing "$\sum 1/n^2 = ?$" at the bottom of the board leads into the next lesson, the topic *Euler, the Basel sum and coprime probability.*

Notes

- The 06-06 board is the continuation of the 05-27 board (the explicit and implicit forms $C = \frac{1-\sqrt{1-4x}}{2x}$, $C^2 - C + 1 = 0$ in its top-right corner). I dated it June 6 because it ends on the Basel question that opens the June 7 lesson; the original filename read "june-67".
- Binary trees with $B = x + xB^2$ (counted by Catalan numbers) appear in the topic *Exponential generating functions and labelled structures*.